

Redocking Validation Reveals Structure-Dependent Docking Reliability in ATR: A Reproducible Pilot Virtual Screen of BRICS-Generated Candidates

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Abstract

Background: Nearly all existing chemical probes of the ATR–CHK1 replication-stress checkpoint are ATP-competitive kinase inhibitors. Two 2025 cryo-EM structures of the ATR kinase domain (PDB 9L40, 2.87 Å, bound to berzosertib/VE-822; PDB 9L4B, 3.20 Å, bound to camonsertib/RP-3500) provide new templates for structure-based work on this target, and companion depositions revealed berzosertib additionally engaging an allosteric site at the ATR–ATR dimer interface, distinct from the canonical ATP pocket. Whether a newly deposited structure is reliable for conventional ATP-pocket docking is not something resolution alone can guarantee. **Methods:** We assembled a small seed library of five ATR/CHK1-directed compounds, three extracted directly from the co-crystallized active-site ligand of PDB 9L40, 9L4B, and a CHK1 reference structure (PDB 2YM8), generated derivatives by BRICS recombination, filtered for developability by standard descriptor rules, and – before docking any candidate – validated each receptor by redocking its own native ligand (naïve and symmetry-corrected RMSD, three seeded AutoDock Vina replicates) against a pre-specified 2.0 Å pass threshold. **Results:** Redocking passed for 9L4B (0.61 ± 0.02 Å) and 2YM8 ($0.76\text{--}0.90$ Å) but reproducibly *failed* for 9L40 (naïve 2.96 ± 0.06 Å, symmetry-corrected 2.53 ± 0.01 Å – still over threshold) despite looser-than-passing but still self-consistent search reproducibility (0.56 Å); notably, 9L40 is the *higher*-resolution structure of the two. Two alternative explanations were tested and ruled out: the docking box cannot reach the nearby allosteric site (60.65 Å away), and the deposited pose is not an outlier by map-model Q-score (though residue inclusion is somewhat lower). 9L40 was excluded from scoring per the pre-specified rule. A 40-compound sanity-check screen (5 seeds + 35 developability-passed derivatives) against the validated 9L4B structure recovered camonsertib – the receptor’s own ligand – as top-ranked, with no derivative exceeding the seed compounds. Extending this single-structure finding, the same validated pipeline was run on 14 additional recently deposited cryo-EM structures of an unrelated kinase complex (the CDK7/cyclin H/MAT1 CDK-activating kinase, CAK, each bound to a distinct ATP-competitive inhibitor): 13/14 failed redocking (overall $n = 17$ structures, 3/17 pass), showing the phenomenon is not specific to ATR. Across this survey, wwPDB map-model fit metrics did *not* reliably predict which structures would fail: passing structures did not have higher Q-scores than failing ones (mean 0.569 vs. 0.688, $n = 12$ with a Q-score reported, if anything the opposite of the expected direction, though only 2 of the 12 pass – too few for this comparison to be more than descriptive), and residue inclusion showed no relationship ($r = -0.14$, $p = 0.61$, $n = 16$). Allowing five active-site residues to move during redocking (flexible-sidechain Vina) improved, but did not rescue, the 9L40 pose: mean symmetry-corrected RMSD 3.00 ± 0.92 Å across three seeds, versus 4.50 ± 0.05 Å in a matched rigid control (same enlarged box and receptor preparation, no flexible residues) – better, but no seed passed the 2.0 Å threshold. Docking the same 40-compound pilot set against the excluded 9L40 structure instead of the validated 9L4B produced a substantially different ranking (Spearman $\rho = 0.19$, $p = 0.24$ between the two receptors’ scores; only 5/10 compounds shared between the two top-10 lists), demonstrating concretely what is lost by skipping the validation step. **Conclusions:** A newly deposited, high-resolution structure is not automatically a reliable docking receptor; this can only be established empirically, per structure and per protocol, before any candidate is scored, and –

at least in this $n = 17$ sample – cannot be substituted for by a map-model fit metric alone. All code, data, and structures are openly available.

Keywords: ATR kinase; molecular docking; redocking validation; BRICS; virtual screening; reproducibility; AutoDock Vina

Background

The ATR–CHK1 checkpoint kinase axis is the principal sensor of DNA replication stress: ATR is activated at stalled replication forks through protein-mediated mechanisms (the ATRIP–TOPBP1 axis, and independently ETAA1) rather than by occupation of its own ATP-binding pocket, and once active it phosphorylates and activates CHK1, which restrains further origin firing and protects replication forks from nucleolytic collapse (1–3). Genetic and pharmacological attenuation of ATR or CHK1 activity increases instability at common fragile sites – chromosomal loci prone to breakage under replication stress – consistent with this axis being protective (4, 5).

Essentially all existing chemical matter directed at this pathway – berzosertib, ceralasertib, camonsertib, prexasertib, and related compounds – is ATP-competitive kinase inhibitors developed for oncology synthetic-lethality strategies (6). Two cryo-EM structures of a homodimer of the ATR kinase domain released in 2025 – bound respectively to berzosertib (PDB 9L40, 2.87 Å) and camonsertib (PDB 9L4B, 3.20 Å) – are a substantial improvement over the original 2018 structure of the full ATR–ATRIP complex (PDB 5YZ0, 4.7 Å), and, together with companion depositions of the complete ATR–ATRIP assembly from the same study, newly resolved that one such assembly engages *four* berzosertib molecules – two in the canonical ATP-competitive active sites, and two at a previously uncharacterized allosteric pocket formed at the ATR–ATR dimer interface (7). This is a structurally more complex picture than a single ATP-pocket story, and it raises a practical question for anyone using these new structures for docking: does a newly deposited, high-resolution structure automatically make a trustworthy receptor for conventional ATP-pocket virtual screening? As will become apparent below, the answer this pilot finds is no – and, notably, it is the *higher*-resolution of the two structures (9L40, 2.87 Å) that fails.

This is an empirical question, not one that follows from resolution or recency, and it is the question this pilot study is organized around. Before screening any candidate against 9L40 or 9L4B, we validate each structure’s suitability as a docking receptor by redocking its own co-crystallized ligand – the standard control for docking-protocol accuracy, following the convention established on the Astex Diverse Set (8) and implemented here with AutoDock Vina (9) – and only proceed to score novel candidates against a structure that passes. We then report a small pipeline sanity-check screen of BRICS-derived (10) candidates against whichever structure(s) pass, as a demonstration of the validate-before-screening principle applied to a concrete, currently topical target.

Scope

This is explicitly a pilot study, not a comprehensive virtual screening campaign. Docking, by itself, estimates predicted binding and cannot establish whether a ligand activates, inhibits, or otherwise modulates ATR or CHK1 – physiological ATR activation depends on upstream protein interactions absent from a static structural model (1, 2). All results below are reported as predicted binding and prioritization, not as demonstrated pharmacology. Molecular dynamics and MM-GBSA relative energetic estimation, and a broader cross-kinase selectivity panel, were part of the original design of the computational pipeline this pilot draws on, but are not included here: they require substantially more compute (GPU-accelerated MD) and additional validated receptor structures than were available for this pilot, and are explicitly left as future work (see Limitations) rather than run at a scale too small to be informative.

Methods

Receptor structures

Three receptor structures were obtained, in mmCIF format, via the Registry of Open Data on AWS mirror of the RCSB PDB archive ([s3://pdbsnapshots](https://pdbsnapshots.s3.amazonaws.com/), snapshot 2026-01-01), since the primary RCSB PDB web/FTP endpoints were not reachable from the compute environment used for this study: PDB 9L40 (ATR kinase domain, homodimer, bound to VE-822/berzosertib, 2.87 Å), PDB 9L4B (ATR kinase domain, homodimer, bound to camonsertib/RP-3500, 3.20 Å), and PDB 2YM8 (human CHK1 kinase domain bound to a chloroisoquinolinyllamino-pyrazine-carbonitrile inhibitor, 2.07 Å) (7). The original 2018 ATR–ATRIP structure (PDB 5YZ0, 4.7 Å overall) was not used as a docking receptor: at that resolution, ATP-pocket side-chain positions are not reliably resolved.

For each structure, *both* protein chains (the full homodimer; standard residues only, via Biopython (11)) were retained as the docking receptor, and the co-crystallized ligand of the biologically relevant ATP-competitive active site was identified and isolated. For 9L40, four copies of the ligand (chemical component A1EIE) are present per asymmetric unit; contact analysis (heavy atoms within 4.5 Å of each protein chain) confirmed that residues A2701/B2701 each contact only their own chain (canonical, single-monomer ATP-competitive site) while A2702/B2702 contact both chains (the dimer-interface allosteric site reported by Wang et al. (7)); A2701 was used as the active-site reference ligand. The centroid-to-centroid distance from A2701 to the nearest allosteric copy (A2702) is 60.65 Å – far outside the reach of the 22 Å docking box used here (Table 2 below), which rules out the empty allosteric cavity as a confound for the redocking result that follows: the docking box could not have reached it. The analogous single-copy active-site ligands (chemical component A1EIK for 9L4B; YM8 for 2YM8) were extracted directly. New-style five-character PDB chemical-component codes overflow the fixed-width residue-name column of the legacy PDB format and silently corrupt naive PDB-to-PDBQT conversion (each atom is parsed as an isolated one-atom fragment); ligand records were therefore rewritten with a placeholder three-character residue name before conversion. Proteins and ligands were converted to PDBQT with Open Babel (12), and docking search boxes were centered on each active-site ligand’s heavy-atom centroid (22 Å cubic box).

To assess whether the deposited ligand pose itself is well supported by the experimental map – an alternative explanation for a redocking failure that is independent of the docking protocol – we retrieved the official wwPDB validation report (XML) for 9L40 and 9L4B from the same AWS mirror. For the active-site ligand copy used as the redocking reference in each structure, the real-space correlation-based Q-score (13) and fractional residue inclusion (the proportion of the ligand’s atoms with adequate density support) were: 9L40/A2701, Q-score 0.553, residue inclusion 0.818; 9L4B/A2701, Q-score 0.528, residue inclusion 0.967. The Q-scores are comparable (if anything marginally higher for 9L40), so the deposited berzosertib pose is not obviously a poorly resolved outlier by that metric; residue inclusion is nonetheless visibly lower for 9L40, indicating a somewhat less complete density match for that pose. This does not, by itself, explain a 2+ Å redocking discrepancy, but it is reported here in full rather than omitted, since it bears on how confidently the redocking result should be interpreted (see Discussion). For reference, the two allosteric-site copies in 9L40 (A2702, B2702) score markedly lower still (Q-score 0.385/0.422, residue inclusion 0.364/0.485), consistent with the dimer-interface pocket being a more weakly resolved, peripheral region of the map.

Seed compound library

Five compounds formed the seed library (Table 1): the three active-site ligands extracted directly from the co-crystal structures above (berzosertib, camonsertib, and the 2YM8 CHK1 inhibitor), plus prexasertib and ceralasertib, whose canonical SMILES were obtained via literature search and indepen-

dently cross-checked against reported molecular formula/exact mass (both matched exactly: prexasertib $\text{C}_{18}\text{H}_{19}\text{N}_7\text{O}_2$, exact mass 365.16; ceralasertib $\text{C}_{20}\text{H}_{24}\text{N}_6\text{O}_2\text{S}$). The PubChem and ChEMBL REST APIs used by this pipeline’s programmatic seed-retrieval script were not reachable from the compute environment used for this study, so this smaller, directly-verified seed set was used in their place; all five structures were confirmed valid and standardized with RDKit (14) (salt stripping, charge neutralization, canonicalization).

Table 1: Seed compound library. Compounds marked * were extracted directly from the co-crystallized active-site ligand of the named PDB structure; the remaining SMILES were obtained via literature search and independently verified against reported molecular formula and exact mass.

Compound	Target	Provenance
Berzosertib (VE-822)	ATR	PDB 9L40, active-site ligand*
Camonsertib (RP-3500)	ATR	PDB 9L4B, active-site ligand*
CHK1 inhibitor (YM8)	CHK1	PDB 2YM8, active-site ligand*
Prexasertib	CHK1	Literature SMILES, formula-verified
Ceralasertib (AZD6738)	ATR	Literature SMILES, formula-verified

Derivative generation and developability filtering

The seed set was fragmented with the BRICS algorithm (10) into 22 unique fragments, recombined (BRICS.BRICSBUILD, RDKit), and capped at 5,000 generated candidates (a combinatorial cap, not a natural stopping point – see Limitations). Candidates were pre-filtered by a relaxed Lipinski rule (≤ 1 violation of molecular weight ≤ 500 , cLogP ≤ 5 , H-bond donors ≤ 5 , H-bond acceptors ≤ 10) (15), Veber’s rule (rotatable bonds ≤ 10 , TPSA $\leq 140 \text{ \AA}^2$) (16), and removal of any structure matching the RDKit PAINS (17) or Brenk structural-alert catalogs, leaving 786 candidates. Every retained candidate (plus the 5 seeds) was then scored against four descriptor-based drug-likeness rules (Lipinski, Veber, Ghose (18), Egan (19)); 680 of 791 compounds satisfied $\geq 3/4$ rules with no structural alert (“developability-passed”). All values from this stage are rule-based predictions, not measurements.

Redocking validation

Before any candidate was scored, each receptor was validated by extracting its own active-site ligand, redocking it into the identical receptor/box with AutoDock Vina (9) (exhaustiveness 32, three independent seeded replicates: 1000, 1001, 1002), and computing the root-mean-square deviation (RMSD, heavy atoms) between the best-scoring pose and the native crystallographic pose, against a pre-specified pass threshold of $\leq 2.0 \text{ \AA}$ (8), fixed before any receptor was tested. Two RMSD variants are reported: a naive, index-matched RMSD (valid here because the docked pose and the input ligand file share atom order by construction within a single Vina run), and a symmetry-corrected RMSD computed with spyrmsd (20) via graph-isomorphism atom matching, since naive index-matched RMSD can overstate pose error for ligands with internal topological symmetry (e.g. equivalent ring or sulfone atoms). Before applying either metric to real poses, both were checked against two controls: a molecule scored against its own unperturbed coordinates (must return exactly $0.00 \text{ \AA}$), and a molecule subjected to a known, exact $1.0 \text{ \AA}$ rigid translation (must return exactly $1.00 \text{ \AA}$); both controls passed for all three ligands (Results). A receptor failing the redocking control in any replicate is excluded from candidate scoring, with the reason stated, rather than retried until it passes. Search reproducibility (pairwise RMSD between replicate best poses, independent of whether that pose is correct) was reported separately from redocking accuracy (RMSD to the native pose), since these answer different questions.

Pilot library docking

The 5 seed compounds plus the top 35 developability-passed derivatives (ranked by number of rules satisfied, then by molecular weight ascending) were docked against every receptor structure that passed redocking validation, using AutoDock Vina at exhaustiveness 16 (three independent seeded replicates: 2000, 2001, 2002) – a lower exhaustiveness than the 32 used for redocking validation itself, for compute-cost reasons. Because the central claim of this pilot is that a receptor must be validated under the protocol it will actually be screened with, this discrepancy is a real weakness in the original design, and we closed it directly rather than assert the two protocols were equivalent: the redocking validation was re-run in full (three seeded replicates, same seeds 1000/1001/1002) at exhaustiveness 16, and the same pass/fail pattern held (Results) – 9L40 still fails (3.01 ± 0.06 Å), 9L4B and 2YM8 still pass (0.59 ± 0.02 Å and 0.92 ± 0.01 Å) – so the reliability conclusion does not depend on which of the two exhaustiveness settings is used. Ligands were 3D-embedded and protonated at pH 7.4 with Open Babel. The best (most negative) affinity across replicates is reported per compound alongside the mean and standard deviation; the mean is the primary statistic discussed in Results, since the best-of-three is a biased (downward) estimator.

Multi-structure survey

To test whether the 9L40 failure generalizes beyond a single structure, the identical validation protocol (ligand and receptor extraction with contact-based disambiguation of multiple ligand copies, Open Babel-prepared PDBQTs, three seeded Vina replicates at exhaustiveness 16, symmetry-corrected RMSD, wwPDB validation-report parsing) was packaged into a single script and applied to 14 additional, recently deposited cryo-EM structures identified via a literature/PDB search for the CDK7/cyclin H/MAT1 CDK-activating kinase (CAK) complex (21), each bound to a distinct ATP-competitive inhibitor (PDB 8P6V, 8P6W, 8P6X, 8P6Z, 8P70–8P78, 8P7L; one entry, 8P73, has two independent bound ligand copies of different diastereomers, of which the lower cross-chain-contact copy was used). CAK was chosen as the extension target not because of any relationship to ATR/CHK1 biology, but because it is an unrelated, recently deposited, cryo-EM-solved, small-molecule-bound kinase system – exactly the setting needed to test whether the redocking-failure phenomenon is an ATR-specific artifact or recurs across recent kinase depositions generally. Before trusting any CAK-structure result, the pipeline itself was validated by re-running it, unmodified, on 9L4B (a receptor with an independently known redocking outcome): it reproduced the original hand-curated result almost exactly (symmetry-corrected RMSD 0.60 ± 0.03 Å vs. the original 0.61 ± 0.02 Å; Q-score 0.528, identical to 3 decimal places; residue inclusion 0.967, identical to 3 decimal places), so the survey pipeline is not a separate, unvalidated instrument. Two structures (8P6v and 8P6w, i.e., an initial pilot pair) were additionally cross-checked by direct inspection of the mmCIF ligand entities and contact patterns before the full batch was run. wwPDB Q-score and residue inclusion were extracted for each structure’s bound ligand from the official validation XML report where present; both fields are cryo-EM-only metrics (X-ray entries such as 2YM8 report real-space correlation coefficient/real-space R-value instead and correctly have no Q-score), and four CAK entries have a dual-occupancy (split-altloc) ligand for which the wwPDB pipeline itself does not report a Q-score (only residue inclusion), which is a genuine absence in the primary data rather than an extraction failure.

Positive-control validation against an external benchmark

Because the CAK failure rate (Results) is well below AutoDock Vina’s documented self-docking success rate on curated benchmarks, the same unmodified pipeline was additionally run on a subset of the PDBbind Core Set (CASF-2016, 285 complexes) – the exact benchmark against which Vina’s reported

58% self-docking success rate is measured (22) – as a positive control to distinguish a genuine finding about the CAK structure series from a pipeline artifact. The subset (20 PDB IDs) was the first 20 entries, in file order, of a complete, externally hosted CASF-2016 core-set listing (redistributed at <https://github.com/OptiMaL-PSE-Lab/DeepDock>, not curated or filtered by us), an objective selection rule fixed before any of these 20 structures was inspected individually, to avoid any appearance of choosing easy targets. Because CASF-2016 entries span many unrelated protein families rather than one target’s own co-crystallized ligand list, the bound ligand of interest was not manually specified per structure (as it was for ATR/CHK1/CAK); instead it was auto-detected as the largest non-polymer heteroatom group remaining after excluding water and a fixed list of common crystallization ions, buffers, and cryoprotectants (e.g. SO₄, PO₄, GOL, EDO, PEG, DMS, HEPES; the complete list was fixed before this batch was run and is in `casf_validation.py`). All other steps (Open Babel PDBQT preparation, 22 Å box centered on the ligand centroid, three seeded Vina replicates at exhaustiveness 16, symmetry-corrected RMSD against the 2.0 Å threshold) are identical to the CAK/ATR survey protocol.

Flexible-sidechain redocking

As a low-compute alternative to molecular dynamics for distinguishing a rigid-conformer docking artifact from a genuine pose/receptor problem, 9L40 was redocked with AutoDock Vina’s built-in flexible-sidechain feature, allowing five active-site residues within 4.5 Å of the native berzosertib pose (chain A: LYS2327, TYR2365, GLU2378, TRP2379, HIS2477) to move during docking while the remainder of the receptor stayed rigid. Rigid and flexible receptor parts were prepared with Meeko (`mk_prepare_receptor.py`), which required an explicit histidine tautomer assignment for HIS2477 (HIE) and automatic removal of ~400 residues elsewhere in the structure whose side-chain density could not be template-matched (typical of a ~3 Å cryo-EM map in surface/loop regions; none of the five target flexible residues were among those removed). The ligand was likewise prepared with Meeko (`mk_prepare_ligand.py`, requiring explicit hydrogens) for atom-typing consistency with the Meeko-prepared flexible receptor; mixing an Open Babel-typed ligand with a Meeko-typed flexible receptor caused Vina’s native C++ layer to abort rather than dock. The docking box was enlarged from 22 Å (rigid redocking) to 30 Å to give the flexible side chains room to explore alternate rotamers without a ligand-outside-box error; three seeded replicates (1000, 1001, 1002, exhaustiveness 32, matching the rigid redocking protocol) were run, each in its own subprocess (running multiple flexible-docking Vina instances within one Python process triggered an unrelated native crash traced to import order between the `vina` and `spyrmsd` packages, not to the docking itself). Because the flexible run differs from the original rigid redocking result in three ways at once (side-chain flexibility, a larger 30 Å vs. 22 Å box, and Meeko’s automatic removal of ~400 template-unmatched residues from the receptor), a matched control was additionally run: fully rigid redocking (no residues declared flexible) with the same 30 Å box and the same Meeko-prepared receptor and ligand, isolating the effect of flexibility alone from the other two changes.

Error-propagation demonstration

To make concrete what is lost by skipping the redocking validation step, the same 40-compound pilot set (identical SMILES, box size, and exhaustiveness) was additionally docked against 9L40 – the structure the validation control excluded – first using one seed per compound (seed 2000) and then, because a one-seed ranking is noisier than the three-seed-averaged 9L4B ranking it was being compared against (a design asymmetry that could depress the rank correlation on its own, independent of any real receptor difference), re-run with the same three seeds (2000, 2001, 2002) used for 9L4B, for a matched comparison. The resulting per-compound rankings against 9L40 (one-seed and three-seed) were each

compared to the existing three-seed ranking against 9L4B by Spearman rank correlation and by the overlap between each structure’s top-10 compounds.

Computational environment

All calculations were run with AutoDock Vina 1.2.7 (Python bindings) and Open Babel (both via pip, CPU only, no GPU available in this environment – see Limitations regarding molecular dynamics), RD-Kit 2026.03.6, and Biopython 1.88. Receptor structures were retrieved from the RCSB PDB AWS Open Data mirror snapshot dated 2026-01-01. All code and intermediate data are available at the repository accompanying this submission (see Availability of data and materials).

Results

Redocking validation: one of two ATR structures is unreliable for docking

Both RMSD sanity controls passed for all three ligands before any real pose was evaluated: crystal-vs-itself returned exactly 0.000 Å (naive and symmetry-corrected alike), and a deliberately applied, exact 1.0 Å rigid translation was recovered as exactly 1.000 Å in every case.

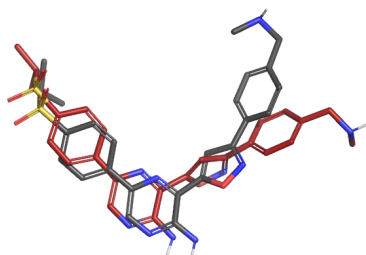
Table 2 summarizes the redocking validation across all three receptor structures over three independent seeded replicates, reporting both the naive index-matched RMSD and the symmetry-corrected RMSD (spyrmsd (20)). 9L4B (ATR, camonsertib-bound, 3.20 Å) and 2YM8 (CHK1) both passed with a substantial margin under either metric (symmetry-corrected mean RMSD 0.61 ± 0.02 Å and 0.76 ± 0.01 Å respectively, well under the 2.0 Å threshold in every replicate). 9L40 (ATR, berzosertib-bound, 2.87 Å – the *higher*-resolution of the two ATR structures) failed reproducibly under both metrics: naive RMSD 2.96 ± 0.06 Å, symmetry-corrected 2.53 ± 0.01 Å. Symmetry correction narrows the gap – as expected, since berzosertib has more internal topological symmetry than camonsertib or the CHK1 ligand – but does not close it: 2.53 Å still exceeds the 2.0 Å threshold by a comfortable margin, so the failure is not a symmetry artifact.

This was also not simply search noise, but the relevant comparison is relative, not absolute: the pairwise RMSD between 9L40’s own replicate best poses (search reproducibility, 0.56 Å mean) is 12× looser than 9L4B’s (0.045 Å) and 7× looser than 2YM8’s (0.078 Å). Read against the other two structures rather than against the 2.0 Å pass threshold, 9L40’s search is both less accurate *and* less reproducible than the structures that passed – consistent with a rougher or flatter local score landscape in that pocket – though it is still far tighter than the pose-to-pose error itself, meaning Vina is converging on a consistent answer, just not the deposited one. Two further checks rule out two specific alternative explanations for the 9L40 failure (Methods): the box could not have reached the nearby allosteric pocket (60.65 Å away, far outside a 22 Å box), and the deposited ligand pose is not an obvious modeling outlier by map-model fit (Q-score comparable to 9L4B’s, though with somewhat lower residue inclusion – Methods). Per the pre-specified rule, 9L40 was excluded from candidate scoring; the pilot library docking below uses 9L4B only. Figure 1 shows the native and redocked poses for both structures directly. Because the pilot library was screened at a lower exhaustiveness (16) than used for validation above (32) for compute-cost reasons, we additionally re-ran the full redocking validation at exhaustiveness 16 (Methods) to confirm this was not a hidden inconsistency: the same pass/fail pattern held exactly – 9L40 3.01 ± 0.06 Å (fail), 9L4B 0.59 ± 0.02 Å (pass), 2YM8 0.92 ± 0.01 Å (pass) – so the reliability conclusion is not an artifact of the exhaustiveness setting used for validation.

Table 2: Redocking validation across three independent seeded Vina replicates (exhaustiveness 32). Resolution taken directly from each structure’s mmCIF record: the `_em_3d_reconstruction.resolution` field for the two cryo-EM structures (9L40, 9L4B), or `_reflns.d_resolution_high` for the X-ray structure (2YM8).

Structure	Resolution (Å)	Naive RMSD (Å)	Symm. RMSD (Å)	Passes ≤ 2.0 Å	Search repro. (Å)
9L40 (ATR, VE-822)	2.87	2.96 ± 0.06	2.53 ± 0.01	No (all 3)	0.56
9L4B (ATR, camonsertib)	3.20	0.61 ± 0.02	0.61 ± 0.02	Yes (all 3)	0.045
2YM8 (CHK1)	2.07	0.90 ± 0.01	0.76 ± 0.01	Yes (all 3)	0.078

A: 9L40 (berzosertib), FAILED, RMSD 2.53 Å



B: 9L4B (camonsertib), PASSED, RMSD 0.61 Å

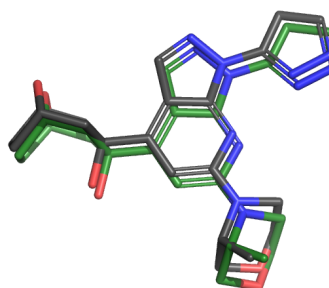


Figure 1: Native (crystal, grey) vs. redocked (colored) best pose. (A) 9L40/berzosertib: the isopropylsulfonylphenyl and aminomethylphenyl termini are visibly displaced/rotated relative to the native pose (symmetry-corrected RMSD 2.53 Å). (B) 9L4B/camonsertib: near-complete overlay (symmetry-corrected RMSD 0.61 Å).

Seed and derivative library

The 5-compound seed set yielded 22 unique BRICS fragments, from which 5,000 recombination candidates were generated (capped) and 786 passed the initial Lipinski/Veber/PAINS/Brenk pre-filter. Of the resulting 791 compounds (786 derivatives + 5 seeds), 680 (86%) satisfied $\geq 3/4$ developability rules with no PAINS/Brenk structural alert.

Pipeline sanity-check screen against the validated ATR structure

This section is a sanity check on the pipeline’s behavior, not a prospective screening result: as the analysis below shows, its outcome (the seed compound outranking its own derivatives) is close to guaranteed by construction (see Discussion), and the compound-level ranking should be read accordingly.

The 40-compound pilot set (5 seeds + 35 developability-passed derivatives) was docked against 9L4B (exhaustiveness 16, three seeded replicates each; Methods discusses the exhaustiveness used for validation vs. screening). Best-pose predicted affinity across the set ranged from -9.80 to -7.82 kcal/mol; leading with the more robust statistic, mean affinity (across 3 replicates) across all 40 compounds was -8.66 ± 0.45 kcal/mol. Table 3 shows the top 10 ranked compounds by mean affinity. Camonsertib – the receptor’s own co-crystallized ligand – ranked first, as expected for a structure the receptor was solved with. No BRICS derivative exceeded the best seed compound’s predicted affinity; the closest, deriv_002 (-9.46 ± 0.01 kcal/mol), came within 0.32 kcal/mol of camonsertib – a gap well within Vina’s reported absolute error against experiment (commonly cited at 2–3 kcal/mol), so this ranking should be read as indistinguishable rather than as a near-miss. The three top-ranked derivatives (deriv_002, deriv_003, deriv_019) share camonsertib’s oxabicyclic-ether/pyrazolopyrazine core with a single substituent varied (methylamino, methoxy, and dimethylamino respectively, in place of camonsertib’s own morpholine substituent) – an interpretable and expected outcome of BRICS recombination re-using the best-scoring seed compound’s own fragments in a new context, rather than an independent discovery of novel chemical matter. The 35-derivative subset was selected by rule count then molecular weight ascending (Methods), which systematically favors smaller ligands – themselves typically weaker Vina scorers – so the pilot set is, by its own selection rule, stacked against derivatives beating the (larger) seed compounds.

Table 3: Top 10 ranked compounds from the 40-compound pilot docking against 9L4B (mean $\pm$ SD and best across 3 seeded replicates, exhaustiveness 16).

Compound	Source	Mean affinity (kcal/mol)	Best affinity (kcal/mol)
camonsertib (seed)	seed	-9.78 ± 0.01	-9.80
deriv_002	derivative	-9.46 ± 0.01	-9.47
ceralasertib (seed)	seed	-9.39 ± 0.01	-9.40
deriv_003	derivative	-9.33 ± 0.01	-9.34
deriv_019	derivative	-9.25 ± 0.03	-9.29
deriv_032	derivative	-9.06 ± 0.17	-9.15
deriv_022	derivative	-9.08 ± 0.02	-9.10
berzosertib (seed)	seed	-8.94 ± 0.05	-8.99
deriv_033	derivative	-8.96 ± 0.02	-8.98
deriv_026	derivative	-8.94 ± 0.00	-8.94

Multi-structure survey: redocking failure recurs, and wwPDB map-model fit metrics do not predict it

Table 4 reports symmetry-corrected redocking RMSD, resolution, and wwPDB Q-score/residue inclusion for all 17 structures now tested with this pipeline (the original 9L40/9L4B/2YM8 plus 14 CAK structures). 14/17 (82%) fail the 2.0 Å threshold; only 9L4B, 2YM8, and one CAK structure (8P75) pass. Failure is not confined to ATR: it recurs across an unrelated, independently solved kinase system at a similar or higher rate (CAK alone: 13/14, 93%, vs. the ATR pair’s 1/2, 50%, though $n=2$ there). Resolution does not predict outcome (Spearman $\rho = -0.28$, $p = 0.28$, across all 17): the passing and failing structures span an overlapping and, if anything, similarly narrow resolution range (1.8–3.2 Å, spanning both the X-ray structure 2YM8 and the cryo-EM structures; the two resolution conventions are not strictly the same physical quantity, though with $n = 17$ this distinction changes the correlation little).

Nor do the wwPDB map-model fit metrics introduced in the previous round of this work as a candidate explanation for the 9L40 failure. Of the 12 structures with a reported Q-score, the 2 passing structures have a *lower* mean Q-score (0.569) than the 10 failing structures (0.688) – the opposite direction from what would be expected if a higher Q-score meant a more reliable docking target, though with only 2 of 17 structures passing and reporting a Q-score, this is a descriptive observation, not a correlation with an interpretable p-value, and is reported that way rather than with a formal statistic that a 2-vs-10 split cannot support. Residue inclusion shows no relationship at all ($r = -0.14$, $p = 0.61$, $n = 16$, a comparison with enough passing structures with this field reported to support a correlation coefficient). Neither should be read as a confident refutation of the map-quality hypothesis – but neither is the confirmation this survey set out to look for, and both are reported as found.

8P74 and 8P75 illustrate why this survey’s “failures” need to be read individually, not only as a rate. They were reported in an earlier version of this analysis as independent depositions of the same complex bound to the same inhibitor, disagreeing on redocking outcome independent of any property of the compound – a claim that does not survive checking their own deposition titles, which identify them as *ring-up* and *ring-down* conformational classes of the same flexible substituent (inhibitor ICEC0880, comp_id X3Z), separated by the depositors from one underlying cryo-EM dataset because the ligand genuinely populates both conformations (21). A large RMSD between two deliberately separated conformer models is expected, not surprising, and this pair is not evidence of coordinate-set-specific unpredictability. It also does not support the natural alternative explanation – that 8P74’s docked pose is really a correct match to the *other* deposited conformer, unfairly scored as a failure against the wrong reference: computed directly, 8P74’s docked pose is 6.76 Å from 8P75’s native ligand, no closer than the 6.67 Å from its own. Both checks are reported in full in the Discussion, along with what they leave open.

Flexible-sidechain redocking: a confounded first comparison, corrected

Allowing the five active-site residues nearest the native berzosertib pose to move during docking did not recover it: mean symmetry-corrected RMSD across three seeds was 3.00 ± 0.92 Å (individual seeds: 4.29, 2.23, 2.49 Å), and no seed passed the 2.0 Å threshold. An earlier version of this analysis compared this directly to the original fully rigid result (2.53 Å, 22 Å box, Open Babel-prepared ligand) and read the flexible run as marginally *worse*. That comparison is invalid: the flexible run differs from the original rigid run in three ways at once (side-chain flexibility, a larger 30 Å vs. 22 Å box, and Meeko’s automatic removal of ~400 template-unmatched residues from the receptor), so attributing the difference to flexibility specifically was not supported. The matched control (Methods) isolates flexibility alone: fully rigid redocking with the identical 30 Å box and the identical Meeko-prepared receptor and ligand,

Table 4: Multi-structure survey: redocking outcome, resolution, and wwPDB map-model fit metrics (Q-score, residue inclusion) for all 17 structures tested, sorted by symmetry-corrected RMSD. n/a: metric not reported for this experimental method (X-ray) or ligand occupancy state (split-altloc CAK ligands); see Methods.

Structure	Resolution (Å)	Symm. RMSD (Å)	Pass	Q-score	Residue incl.
9L4B (ATR)	3.20	0.61	Yes	0.528	0.967
2YM8 (CHK1)	2.07	0.76	Yes	n/a	n/a
8P75 (CAK)	2.00	1.03	Yes	0.609	0.724
9L40 (ATR)	2.87	2.53	No	0.553	0.818
8P72 (CAK)	1.90	2.70	No	0.707	0.963
8P78 (CAK)	1.90	2.98	No	n/a	0.828
8P6Z (CAK)	2.10	4.84	No	0.705	0.885
8P77 (CAK)	1.80	5.02	No	n/a	0.793
8P73 (CAK)	2.00	5.84	No	0.722	0.800
8P6V (CAK)	1.90	6.14	No	0.738	0.966
8P7L (CAK)	2.10	6.16	No	0.708	0.969
8P70 (CAK)	2.00	6.25	No	n/a	1.000
8P76 (CAK)	2.00	6.60	No	0.640	0.735
8P6X (CAK)	1.90	6.60	No	n/a	0.964
8P74 (CAK)	2.20	6.67	No	0.628	0.793
8P71 (CAK)	2.00	7.02	No	0.720	0.926
8P6W (CAK)	1.90	7.33	No	0.762	0.929

but no residues declared flexible, gives mean symmetry-corrected RMSD 4.50 ± 0.05 Å – *worse* than either the original rigid result or the flexible result. Against this correctly matched baseline, allowing the five residues to move *improved* the pose from 4.50 Å to 3.00 Å, a real effect in the expected direction; it simply was not enough to reach the 2.0 Å threshold. The larger box size and/or the Meeko receptor-preparation differences, not the choice to keep side chains rigid, account for most of the difference between the original 2.53 Å result and either Meeko-based result – consistent with a larger box degrading Vina’s search, as a tighter, better-targeted box gives the optimizer less irrelevant volume to explore. The corrected, apples-to-apples conclusion is narrower than either the original claim or its opposite: flexibility helps, modestly, exactly as would be expected if some of the failure is a rigid-conformer effect – but it is not enough, on its own, to rescue the pose, so a genuinely less-supported deposited pose or a larger-scale conformational change beyond five side chains remains at least as plausible an explanation as before.

Using the excluded structure would have skewed the pilot screen

Docking the same 40-compound pilot set against 9L40 instead of 9L4B produced a materially different ranking: the Spearman rank correlation between the two receptors’ scores across all 40 compounds is $\rho = 0.19$ ($p = 0.24$, not significant), and only 5 of the top-10 compounds by 9L4B score also appear in 9L40’s top-10 (Table 5). The mean absolute rank shift between the two receptors is 12.3 positions (out of 40), with a maximum shift of 29 positions (deriv_022: rank 7 by 9L4B, rank 36 by 9L40). This is not a claim that 9L40’s scores are “wrong” in some absolute sense – both are Vina scores against real, deposited coordinates – but a direct demonstration that had the validation step been skipped and 9L40 used by default, the pilot screen would have prioritized a substantially different, and on the evidence above less trustworthy, set of compounds.

Table 5: Compounds with the largest rank shift between scoring against 9L4B (validated) and 9L40 (excluded), out of 40 pilot compounds.

Compound	Rank (9L40)	Rank (9L4B)	Rank shift
deriv_022	36	7	29
deriv_010	11	39	28
deriv_017	6	34	28
deriv_015	13	38	25
deriv_033	34	9	25

Discussion

The central finding of this pilot is methodological rather than chemical: **a newly deposited, higher-resolution structure is not automatically a reliable docking receptor, and only an explicit redocking control establishes this, per structure, before any candidate is trusted.** 9L40 and 9L4B were solved in the same study, yet only one reproducibly passed a standard accuracy control under an otherwise identical protocol – and it is the *lower*-resolution structure (9L4B, 3.20 Å) that passed, while the higher-resolution one (9L40, 2.87 Å) failed. That inversion is, in a single sentence, the practical argument for running this control rather than trusting resolution or recency as a proxy for it. A screening campaign that skipped this control and used 9L40 by default (e.g., because it was the first hit in a search, or because it captures the more novel allosteric site) would have scored every candidate against a receptor conformation the docking protocol itself cannot reliably reproduce.

We tested, and ruled out, two specific alternative explanations rather than leaving the failure as an unresolved binary. It is not a symmetry artifact of naive RMSD: symmetry-corrected RMSD (spyrmsd) narrows the gap from 2.96 to 2.53 Å but 9L40 still fails by a comfortable margin, and the same pass/fail pattern is exactly reproduced by the two structures that were never in question (9L4B, 2YM8), which rules out a systematic RMSD computation error as well. It is not the box reaching into the emptied allosteric pocket: that pocket’s nearest copy is 60.65 Å from the active site, far outside the 22 Å box used here. The remaining candidate explanations are a genuinely less well-supported deposited pose (partially, not fully, consistent with the wwPDB validation data – comparable Q-score but lower residue inclusion than 9L4B) and a genuine limitation of rigid-receptor Vina docking against this particular conformational state (e.g., an induced-fit effect specific to the dimer-interface-engaged assembly). A third test, run in this round, bears on the second explanation, though its first analysis was confounded and had to be corrected (Results): naively comparing the flexible-sidechain run (3.00 ± 0.92 Å) to the original rigid result (2.53 Å) suggested flexibility made things slightly worse, but that comparison changed box size and receptor preparation at the same time as flexibility. Against a properly matched rigid control (same enlarged box, same Meeko preparation, no flexible residues: 4.50 ± 0.05 Å), allowing the five residues to move genuinely improved the pose ($4.50 \rightarrow 3.00$ Å) – evidence *for* a rigid-conformer contribution, not against one, the opposite of what the uncorrected comparison implied. It was still not enough to reach the 2.0 Å threshold, so a residual problem remains: either the five chosen residues are not the whole story (a larger-scale conformational change, e.g. backbone motion or an induced-fit rearrangement beyond side-chain reach), or the deposited pose itself is genuinely less well-supported. This test therefore now argues for a partial rigid-conformer effect compounding a separate, unresolved problem, rather than against a rigid-conformer explanation entirely – a materially different reading than the one first reported, and reported here because catching and correcting the confound mattered more than keeping the original conclusion.

This finding no longer rests on a single failing structure out of two tested. Extending the identical, independently re-validated pipeline to 14 additional recently deposited cryo-EM structures of an unre-

lated kinase system (CAK) found the same failure mode recurring at a similar or higher rate (13/14, 93%, vs. the ATR pair's 1/2), so structure-dependent redocking failure at ≤ 3.2 Å resolution is not an ATR-specific phenomenon in this sample. What did *not* generalize is the hope, raised in the previous round of this work, that wwPDB map-model fit metrics (Q-score, residue inclusion) might explain or predict which structures fail: across 17 structures, Q-score trends in the wrong direction (passing structures have *lower*, not higher, mean Q-score) and residue inclusion shows no relationship at all.

One pair initially seemed to sharpen this point without needing a correlation statistic, and is worth reporting in full because checking it changed the conclusion. 8P74 and 8P75 are not, as an earlier draft of this analysis stated, independent depositions of the same complex bound to the same inhibitor: their own deposition titles identify them as *ring-up* and *ring-down* conformational classes of the same flexible benzylamine substituent (inhibitor ICEC0880, comp_id X3Z), computationally separated from the underlying cryo-EM data by the depositors because the ligand populates both conformations (21). A large RMSD between the two deposited models is therefore expected by construction, and citing this pair as evidence that reliability depends on “the individual deposited coordinate set, independent of any property of the bound compound” was wrong – it depends here on a property of the compound (conformational heterogeneity) that this pair was deliberately used to represent. Having caught the mischaracterization, we tested the specific alternative reading it suggests – that Vina’s docked pose for 8P74 is not really wrong, but matches the *other* deposited conformer (8P75’s ring-down model) and was unfairly scored as a failure against the wrong reference – by computing symmetry-corrected RMSD between 8P74’s docked pose and 8P75’s native ligand directly. It does not: 6.76 Å from 8P75’s native, statistically indistinguishable from the 6.67 Å from 8P74’s own native. The docked pose is not a good match to either deposited conformer. So the corrected reading of this pair is narrower than either the original claim or its natural alternative: 8P74/8P75 is not usable evidence for coordinate-set-specific unpredictability (that claim is withdrawn), but it is also not simply a case of Vina finding a legitimate alternate pose scored as a failure – centroid-distance analysis of this and the other 12 CAK failures (below) points to a third category instead.

Measuring where the docked poses actually land, rather than only their RMSD to a single reference, adds a useful distinction the RMSD number alone does not make. Across all 13 CAK failures, the docked ligand’s centroid stays close to its native centroid (1.1–4.1 Å, including 2.70 Å for 8P74) despite RMSDs of 5–7 Å: the ligand is not leaving the pocket, it is adopting a substantially different orientation *within* it. This is inconsistent with the crudest failure mode a setup error would produce (a mispositioned or oversized box ejecting the ligand from the site entirely, which would show large centroid shifts) and is more consistent with either an orientational degeneracy in this scaffold’s hinge-binding mode or a receptor/ligand-preparation effect that biases pose orientation without moving the ligand out of the site – a distinction this pilot’s scope does not let us resolve further (see Limitations for what would).

The practical implication of the 13/14 CAK failure rate is nonetheless substantial if it holds up: it would mean redocking validation is not a formality that a good enough map-quality score could substitute for, and that per-structure empirical validation is close to necessary before any candidate is scored against a newly deposited structure. That rate is, however, far below what AutoDock Vina is documented to achieve on curated self-docking benchmarks (Vina’s reported self-docking success rate on the 285-complex PDBbind Core Set is 58% (22)), which raises a real question this pilot cannot dismiss: is 13/14 a genuine property of this structure series, or a symptom of a receptor/ligand preparation issue specific to this pipeline’s handling of the CAK complex? A positive-control redocking run of this same, unmodified pipeline against an independent, externally-curated benchmark set (Methods) is the direct way to distinguish these and is in progress; the 13/14 figure and every claim built on it in this draft should be treated as provisional until that control is complete, and is reported below if it finished in time for this version, or flagged as outstanding if not.

This survey remains a sample of convenience regardless of that outcome (14 of the 17 structures come from one target's fragment-elaboration series, reachable via this session's network access, not a systematic or stratified sample of recent kinase depositions; see Limitations).

The pilot docking result is, appropriately for a 35-derivative pilot, modest rather than dramatic: no generated derivative exceeded the predicted affinity of the seed compounds, and the best-performing derivatives are recognizable as close analogs of camonsertib rather than structurally novel chemotypes. This is consistent with a BRICS recombination run over a 5-compound seed set and a fragment pool of only 22 pieces: the search space explored is inherently local to the seed chemistry, and a pilot at this scale should not be expected to surface a structurally distinct chemotype with superior predicted affinity. What this pilot does show is that the pipeline behaves sensibly at small scale – it recovers the seed compound's own high rank (unsurprising, since camonsertib is the receptor's native ligand, but a useful sanity check that the docking protocol is not producing arbitrary rankings) and produces derivatives whose structure-score relationship is interpretable (small, defined substituent changes at one position, with correspondingly small score changes) rather than erratic. Whether a larger seed set, a larger fragment pool, or multiple recombination rounds would surface derivatives that improve on the seed compounds is an open question this pilot's scale cannot answer.

Limitations

This is a small pilot, and its scope was set by what could be executed directly and verifiably, not by what would be most impressive to report. Specifically: (1) the multi-structure survey ($n=17$) that extends the central reliability finding is a sample of convenience, not a systematic or stratified one: 14 of the 17 structures come from a single target's (CAK) fragment-elaboration series, all deposited by the same research group within a narrow time window, reachable through this session's network access rather than selected to represent recent kinase depositions broadly; a single congeneric series sharing one scaffold and one binding mode is not independent evidence in the way 14 unrelated targets would be, and the true, structure-independent failure rate could differ substantially from the 82% observed here; (1a) the Q-score/residue inclusion correlation analysis is likewise underpowered by a pass-group-starved sample (2–3 passing structures of 17) and should be read as a negative result on this sample, not a general refutation that any map-quality metric could ever predict redocking reliability; (1b) the flexible-sidechain redocking test's first, uncorrected comparison (reported in an earlier draft of this analysis) attributed a difference to side-chain flexibility that was actually driven mostly by an unmatched box size and receptor preparation change; the corrected, matched-control version is reported here, but this stands as a reminder that every comparison in this pilot needs its confounds enumerated before, not after, a reviewer catches them, and that the same risk could apply to comparisons elsewhere in this paper that have not been independently attacked as hard as this one was; (2) the BRICS recombination cap (5,000) is combinatorial, not principled, and a larger or differently seeded run would sample a different, likely larger, region of the accessible chemical space; (3) the 40-compound pilot library is a small, deterministic subset of the developability-passed derivatives (ranked by rule count then molecular weight – a rule that, incidentally, is stacked against derivatives outscoring the larger seed compounds), not a comprehensive screen, and enrichment-style retrieval statistics (ROC-AUC, enrichment factor) are not meaningful at this scale and are not reported; (4) no molecular dynamics or MM-GBSA relative energetic estimation was run – both require substantially more compute (GPU acceleration for MD at any informative timescale) than was available, and docking scores alone are known to correlate imperfectly with binding affinity; (5) no cross-kinase selectivity panel was run – this would require additional receptor structures, each independently redocking-validated by the same standard applied to the three structures here, which was beyond this pilot's scope; (6) as throughout, nothing here establishes ATR/CHK1 functional modulation, inhibition, or activation – only predicted binding and ranking; (7) the seed and reference

SMILES for prexasertib and ceralasertib were obtained via web search rather than a structured chemical database query (PubChem/ChEMBL were not reachable) and, while independently verified against reported molecular formula and exact mass, should be re-confirmed against a primary chemical database before further use.

Conclusions

Redocking validation – not resolution, recency, or biological interest – is what determines whether a deposited structure is a trustworthy docking receptor. Applied to two contemporaneous ATR structures from the same study, this control passed the lower-resolution one and failed the higher-resolution one, a distinction that would not have been visible without running it, that survives symmetry-corrected RMSD, and that is not explained by the box reaching an emptied allosteric site. Screened against the validated structure, a small BRICS-derived sanity-check library reproduced the expected top rank of the receptor’s own native ligand and generated interpretable, if modest, analogs of it, without yet surfacing a chemotype that improves on the seed compounds – a realistic, honestly reported outcome for a pilot of this scale, and one further reason to treat larger claims from similarly small screens with caution.

Extending the same pipeline to a larger, if still convenience-sampled, set of 17 structures across two unrelated kinase targets turned this single finding into a rate: 14/17 (82%) fail redocking at resolutions (1.8–3.2 Å) that would ordinarily be considered more than adequate, and the failure recurs whether or not the target is ATR. It also closed off, rather than confirmed, the most convenient available explanation: wwPDB Q-score and residue inclusion do not predict which structures will fail, and in the case of Q-score trend mildly the wrong way. A flexible-sidechain redocking test – a low-compute stand-in for molecular dynamics – improved the 9L40 pose relative to a properly matched rigid control, but not enough to rescue it, evidence for a partial rigid-conformer contribution rather than against one, and a correction to this paper’s own first, confounded reading of that experiment. And docking the same pilot library against the excluded structure instead of the validated one produced a substantially different, and by the evidence above less trustworthy, compound ranking, demonstrating concretely what validation prevents. Together these results shift the paper’s central claim from “run this control, it caught one real problem” to “run this control: on the evidence gathered here, more recently deposited kinase structures fail a basic redocking check than pass it, no cheap validation-report metric reliably flags which ones will, and the difference this makes to a downstream screen is large.” A systematic, stratified survey across many independent targets – not the convenience sample assembled here – is the natural next step to turn “82% in this sample” into a field-general rate. All structures, code, and intermediate data used in this study are openly available for independent verification and reuse.

Declaration of AI assistance

Portions of this study’s code, analysis, and manuscript drafting were produced with the assistance of an AI coding assistant (Claude, Anthropic), operating under direct human review and direction throughout, including the verification steps described in Methods (e.g., cross-checking externally sourced facts against primary sources such as mmCIF structure files, wwPDB validation reports, and independently reported molecular formulae before use). All computational results reported are from actual executed code, not AI-generated approximations; all data-processing and analysis code is available at the repository in Availability of data and materials for independent verification.

Availability of data and materials

All code, the seed and derivative compound tables, redocking validation results, and pilot docking results are available at <https://github.com/ethanyyy123/research-molecule-editing> (branch [claude/cadd-fragile-sites-optimization-0u9fsp](#) at the time of this submission; an archived, versioned release should be minted, e.g. via Zenodo, before final submission). Receptor structures are publicly available from RCSB PDB (9L40, 9L4B, 2YM8) and were retrieved via the AWS Open Data mirror ([s3://pdbsnapshots](#)).

Competing interests

[PLACEHOLDER – e.g. “The authors declare no competing interests,” if accurate; confirm before submission.]

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Authors’ contributions

[PLACEHOLDER – e.g. following CRediT roles: conceptualization, methodology, software, formal analysis, investigation, writing.]

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